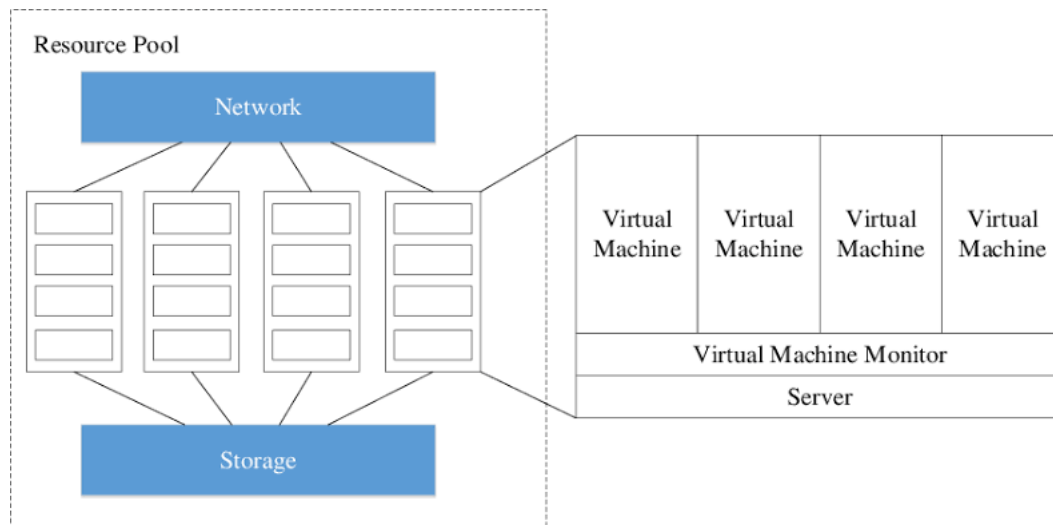


Cloud computing is the on-demand availability of computer system resources, especially data storage and computing power, without direct active management by the user. Large clouds often have functions distributed over multiple locations, each of which is a data center. **example?** Today, there are several examples of cloud computing applications used by both businesses and individuals. One type of cloud service would be streaming platforms for audio or video, where the actual media files are stored remotely. Another would be data storage platforms like Google Drive, Dropbox, OneDrive, or Box. **used?** Cloud Computing **allows you to access resources, data, services, and applications from anywhere you want**, as long as you are connected to the internet. If you are not connected to the internet, some tools and techniques will allow you to access the cloud whenever needed.



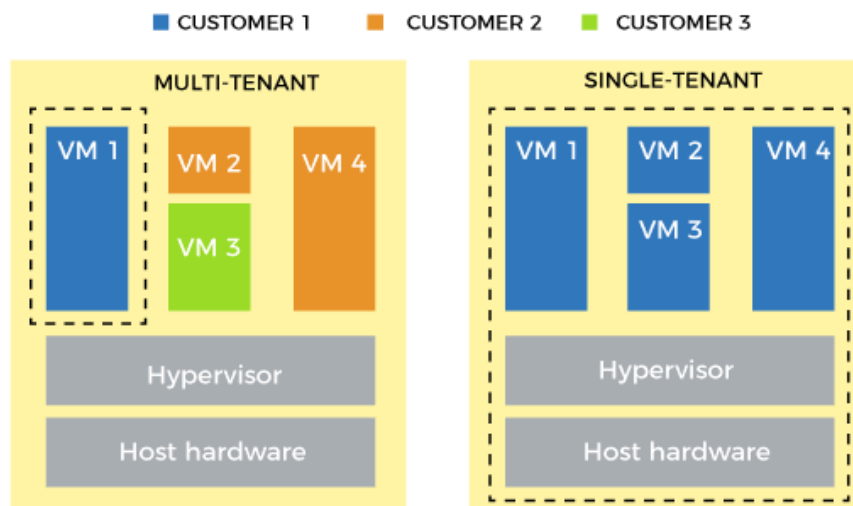
There are **four cloud deployment models: public, private, community, and hybrid**. Each deployment model is defined according to where the infrastructure for the environment is located. There are three main cloud service models: Software as a Service, Platform as a Service, and Infrastructure as a Service. There are many different models to choose from in regard to cloud deployments. Your cloud infrastructure and placement of each workload depends on your business needs. Each cloud deployment model has its own pros and cons.

Resource sharing enables cloud providers and customers to reduce their capital expenditure, operational expenditure, and total cost of ownership. Multiple cloud service providers connect to form a federation to share their resources among themselves to meet the dynamic demand of their customers. Resource Sharing” means **making the collections of one library available to the patrons of. another library efficiently and effectively.** Resource sharing encompasses the technical capabilities, staff expertise and policies necessary to achieve that objective.



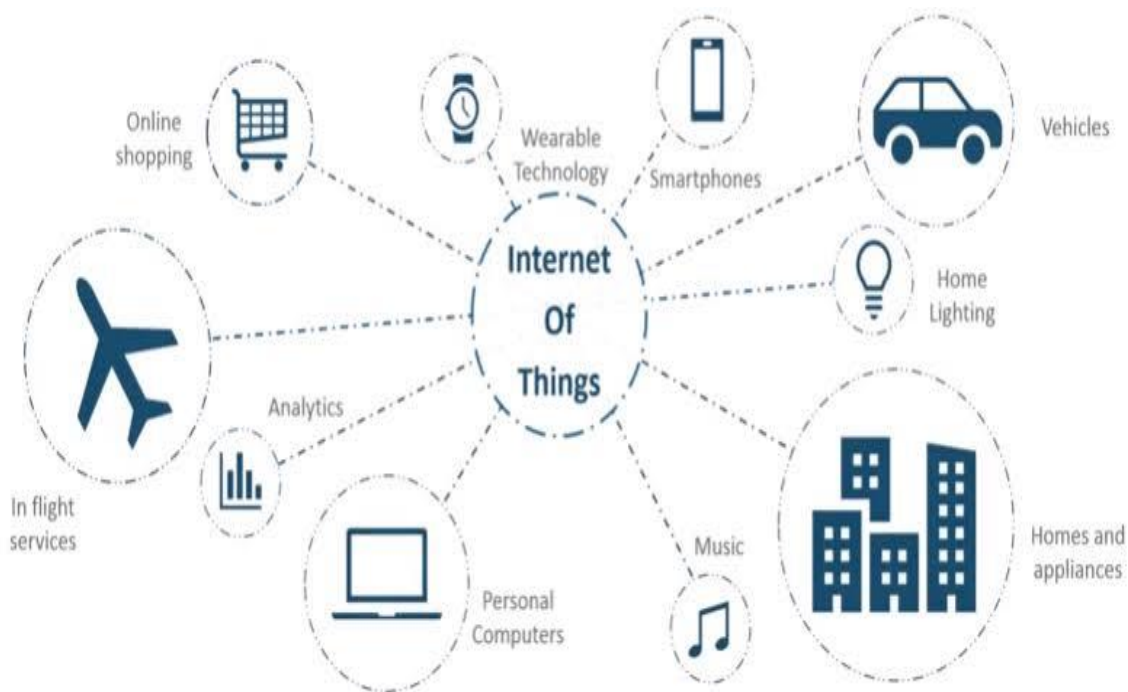
Single-tenant cloud architecture is **one where a single software instance and its supporting infrastructure/database serve only one customer**. In a single-tenant environment, all customer data and interactions are separate from every other customer. Single-tenancy is **an architecture in which a single instance of a software application and supporting infrastructure serves one customer**. Single-tenancy is commonly implemented in software-as-a-service (SaaS) delivery models or in cloud services.

Multi-tenant vs Single-tenant



In cloud computing, **multitenancy** means that **multiple customers of a cloud vendor are using the same computing resources**. Despite the fact that they share resources, cloud customers are not aware of each other, and their data is kept totally separate. Multitenancy is a **software architecture where a single software instance can serve multiple, distinct user groups**. Software-as-a-service (SaaS) offerings are an example of multitenant architecture. **advantages?** Multitenancy allows multiple instances of the given application to operate in a shared environment. Thus, a single instance of software runs on a server and accommodates numerous tenants. Tenants integrate physically, but they're separated logically.

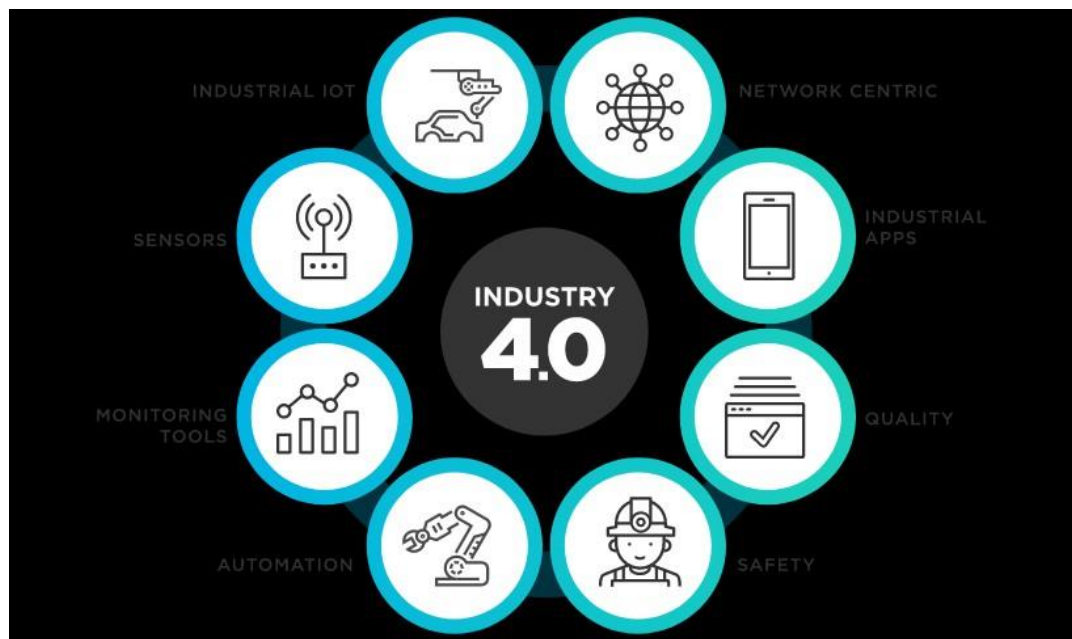
The **Internet of things** describes physical objects with sensors, processing ability, software and other technologies that connect and exchange data with other devices and systems over the Internet or other communications networks. There are two types of IoT: **CIoT and IIoT**. The differences between CIoT and IIoT are: CIoT often focuses on convenience for individual customers, whereas IIoT is strongly focused on the industry sector, improving the efficiency, security, and output of operations with a focus on Return on Investment (ROI). **examples of Internet of things?** In short, the Internet of Things refers to the rapidly growing network of connected objects that are able to collect and exchange data in real time using embedded sensors. **Thermostats, cars, lights, refrigerators, and more appliances** can all be connected to the IoT.



- **Characteristics of the Internet of Things**-Connectivity – ...
- Intelligence and Identity – ...

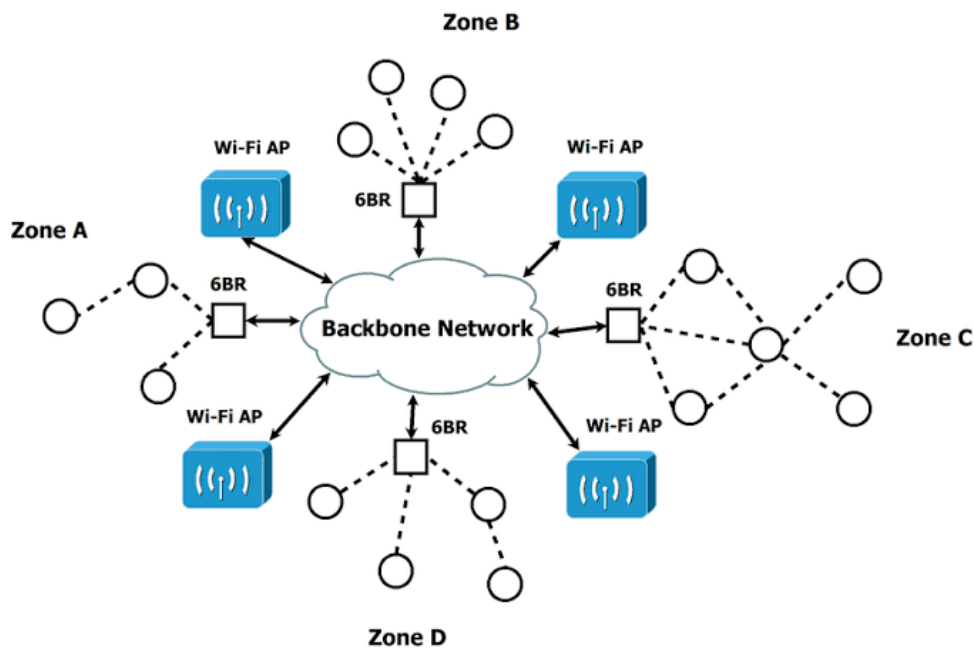
- Scalability – ...
- Dynamic and Self-Adapting (Complexity) – ...
- Architecture –

Industrial IoT, or the Industrial Internet of Things (IIoT), is a vital element of Industry 4.0. **IIoT harnesses the power of smart machines and real-time analysis to make better use of the data that industrial machines have been churning out for years.** The principal driver of IIoT is smart machines, for two reasons. The industrial internet of things (IIoT) is **the use of smart sensors and actuators to enhance manufacturing and industrial processes.**



Infrastructure IoT is a vast term and the required component for Internet of Things infrastructure that varies from industry to industry. Moreover, it also depends on several factors such as sensors and measurements,

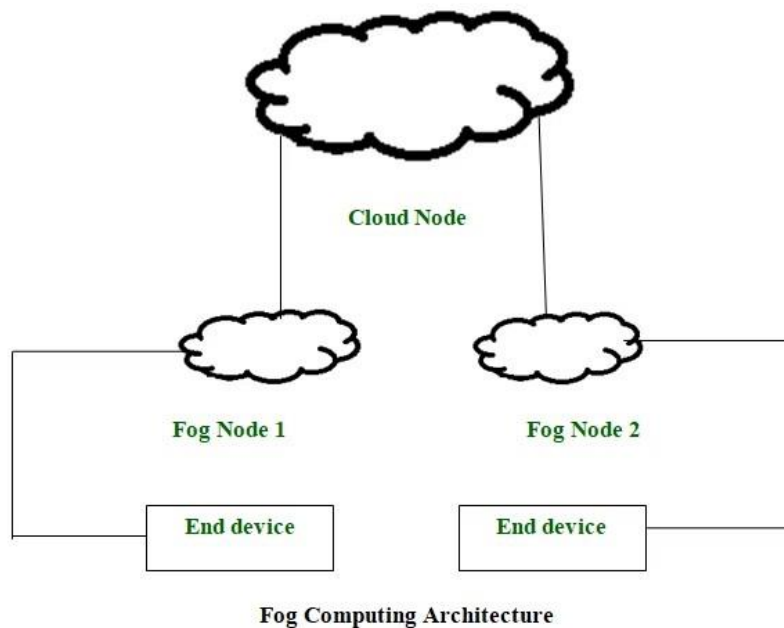
communication protocols, data volume, and analytics. examples of IoT based applications in today's world.



- #1) IoT Sensors. [image source] ...
- #2) IoT Data Analytics. ...
- #4) IoT Connected Factory. ...
- #5) Smart Supply Chain Management. ...
- #6) Smart Barcode Readers. ...
- #7) Smart Grids. ...
- #8) Connected HealthCare System. ...
- #9) Smart Farming.

What Is the Internet of Military Things (IoMT) The Internet of Military Things (IoMT) and the Internet of Battlefield Things (IoBT) are **networks of sensors, wearables, and IoT devices that use cloud and edge computing to increase military capabilities and safety.** **How is IoT used in military?** IoT enables armed forces to survey the battlefield with unmanned aerial drones that are equipped with cameras and sensors. These drones can capture live images, trace the landscape and location of the enemies, and send real-time data to the command center.

Fog computing or fog networking, also known as fogging, is an architecture that uses edge devices to carry out a substantial amount of computation, storage, and communication locally and routed over the Internet backbone. Fog computing is **a computing architecture in which a series of nodes**

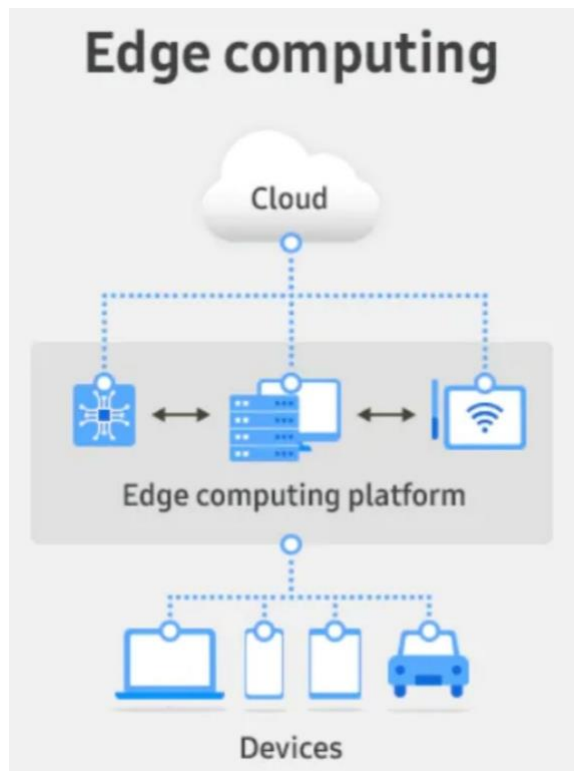


receives data from IoT devices in real time. These nodes perform real-time processing of the data that they receive, with millisecond response time. The nodes periodically send analytical summary information to the cloud. **example?** The most prevalent example of fog computing is perhaps **video surveillance**, given that continuous streams of videos are large and cumbersome to transfer across networks. The nature of the involved data results in latency problems and network challenges. Costs also tend to be high for storing media content.

Edge computing, a strategy for computing on location where data is collected or used, **allows IoT data to be gathered and processed at the edge, rather than sending the data back to a datacenter or cloud.** Together, IoT and edge computing are a powerful way to rapidly analyze data in real-time. Edge computing is already in use all around us – from the wearable on your wrist to the computers parsing intersection traffic flow. Other **examples** include **smart utility grid analysis, safety monitoring of oil rigs, streaming video optimization, and drone-enabled crop management.** **used?** Edge computing is often used to support predictive maintenance efforts, energy efficiency initiatives, custom production runs, smart manufacturing and intelligent operations. **4 Types of Edge Computing – Broadly Categorized**

- Internet of Things Edge.
- On Premise Edge.
- Access Edge.

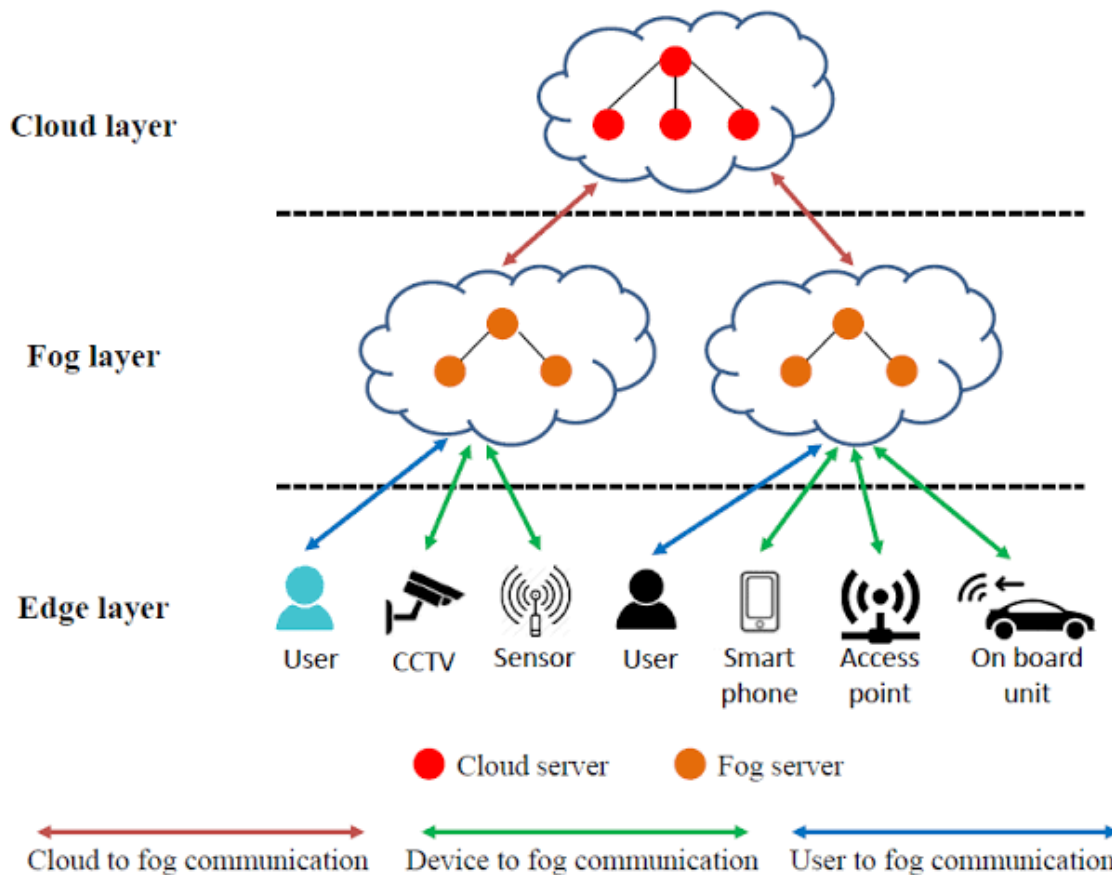
- Network Edge.



Fog computing features three layers: end devices, fog nodes, and clouds.

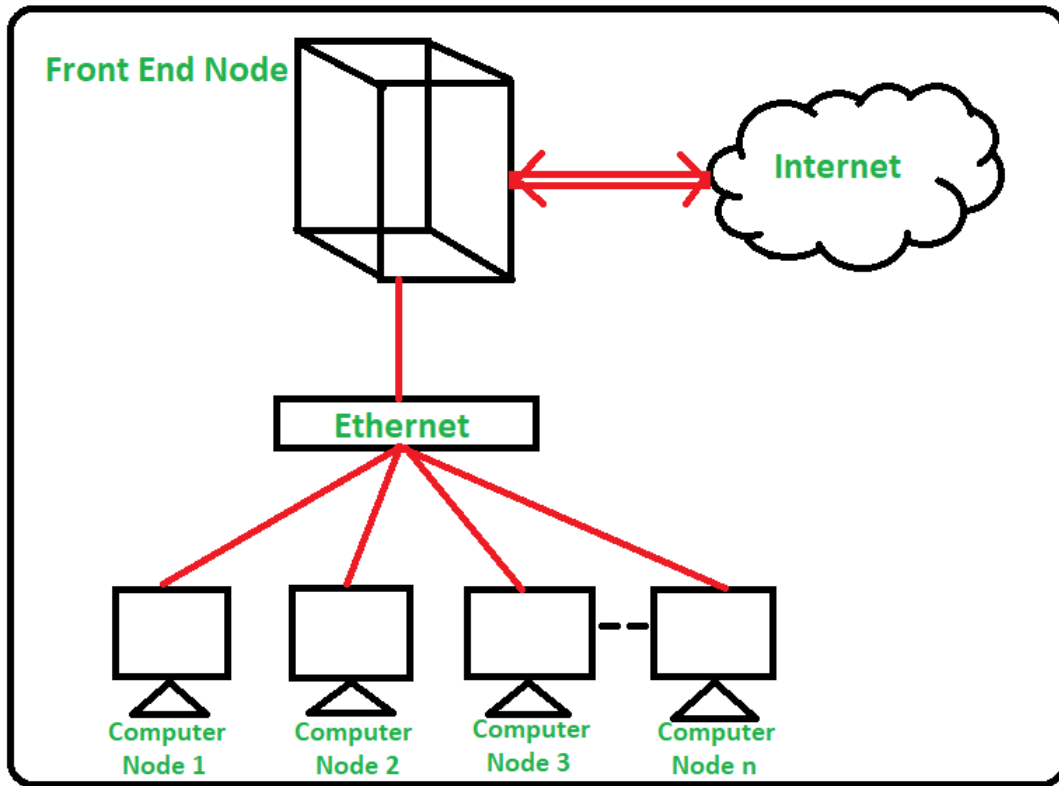
End devices have small capacity but they are present in huge numbers. Fog nodes offer higher, but still limited capacity, and they are less numerous.

Finally, clouds offer large capacity, while their number is low. **Diagram-**



Cluster Computing :

Cluster computing refers to the process of sharing the computation task to multiple computers of the cluster. The number of computers are connected on a network and they perform a single task by forming a Cluster of computers where the process of computing is called as cluster computing.



CHARACTERISTICS OF CLUSTER COMPUTING

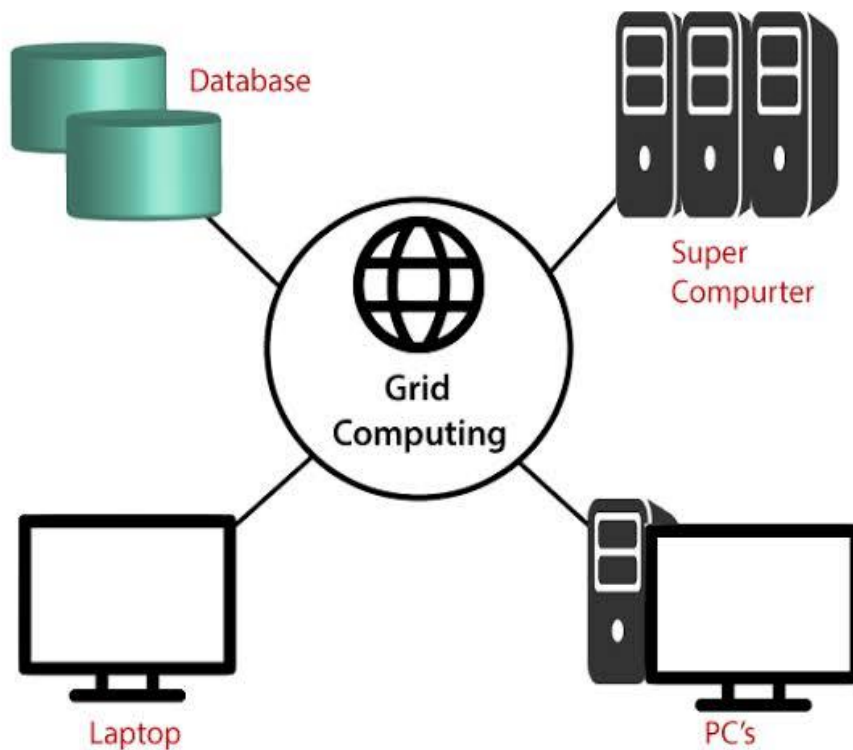
- Tightly coupled computers.
- Single system image.
- Centralized job management and scheduling system.
- On-Demand Self service
- Broad network access
- Resource pooling
- Rapid elasticity

characteristics of cloud computing

- On-demand self-service. ...
- Resource pooling. ...
- Scalability and rapid elasticity. ...
- Pay-per-use pricing. ...
- Measured service. ...
- Resiliency and availability. ...
- Security. ...

- Broad network access.

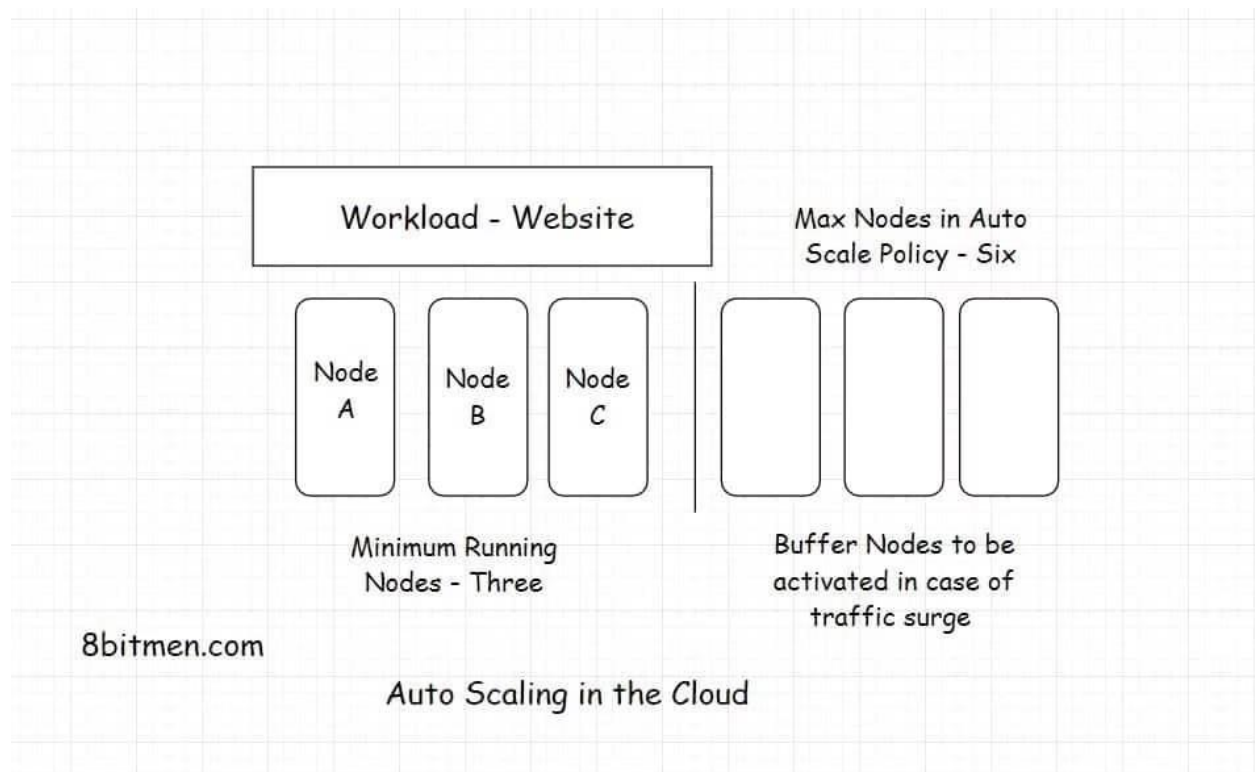
Grid computing is a group of networked computers that work together as a virtual supercomputer to perform large tasks, such as analyzing huge sets of data or weather modeling. All unused resources on multiple computers are pooled together and made available for a single task. Organizations use grid computing to perform large tasks or solve complex problems that are difficult to do on a single computer. For example, **meteorologists use grid computing for weather modeling**.



features of grid computing? Data grid computing allows several users to simultaneously access, change, or transfer distributed data. For instance, a data grid can be used as a large data store where each website stores its own data on the grid. Here, the grid enables coordinated data sharing across all grid users.

What is Auto Scaling?

Auto-scaling is the cloud platform's ability to react to the variation in the live traffic load on the application/workload by spinning up or down server instances on the fly. The ability to auto scale enables the cloud to add or remove additional computing power to the instance cluster based on the demand & thus ensuring smooth handling of the traffic surge/slump. So, when additional instances are added on the fly, they share the traffic load of the existing running servers, diminishing the risk of them crumbling under the heavy traffic load. When the traffic subsides & the workload needs comparatively less computing power the cloud auto-scaler obliges by freeing instances by shutting them down.



Types of autoscaling-Fundamentally there are three types of autoscaling:

- **Reactive.**
- **Predictive.**
- **Scheduled.**

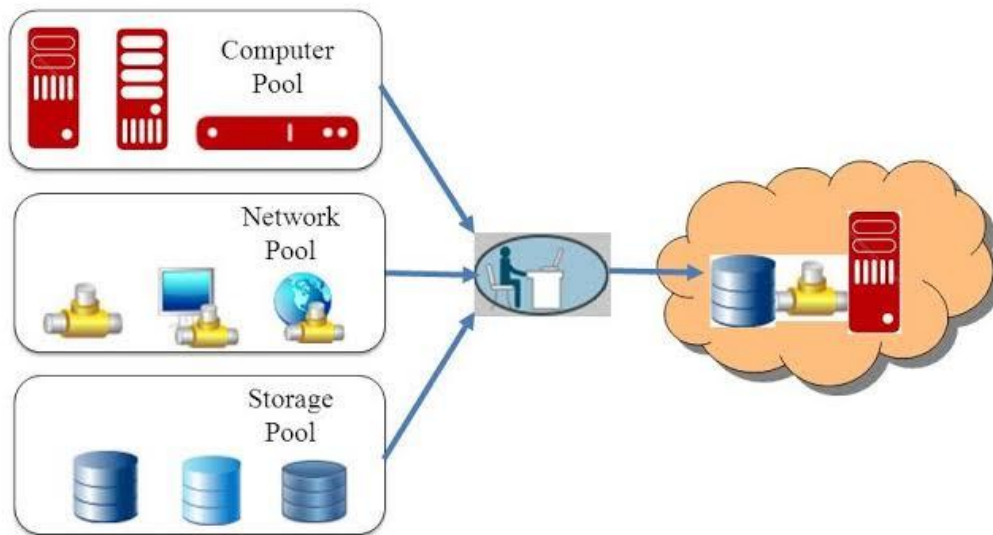
Benefits of autoscalingThere are many benefits that come with using an autoscaling technology or service, rather than having a statically configured instance type in a deployment that does not automatically scale. These benefits include:

- **Lower cost.**
- **Automation.**
- **Reliable performance levels.**

- Improved fault tolerance.
- Service availability.

Resource pooling is when cloud providers offer provisional and scalable services to multiple clients or customers. In other words, space and resources are pooled to serve multiple clients at one time. Depending on a client's resource consumption, usage can be set to provide more or less at any given time. **example-** The kinds of services that can apply to a resource pooling strategy include **data storage services, processing services and bandwidth provided services**. The main **benefits** of resource pooling are, in order, (1) **increased robustness against component failures**; (2) better ability to handle localized surges in traffic; (3) maximized utilization.

Resource Pooling



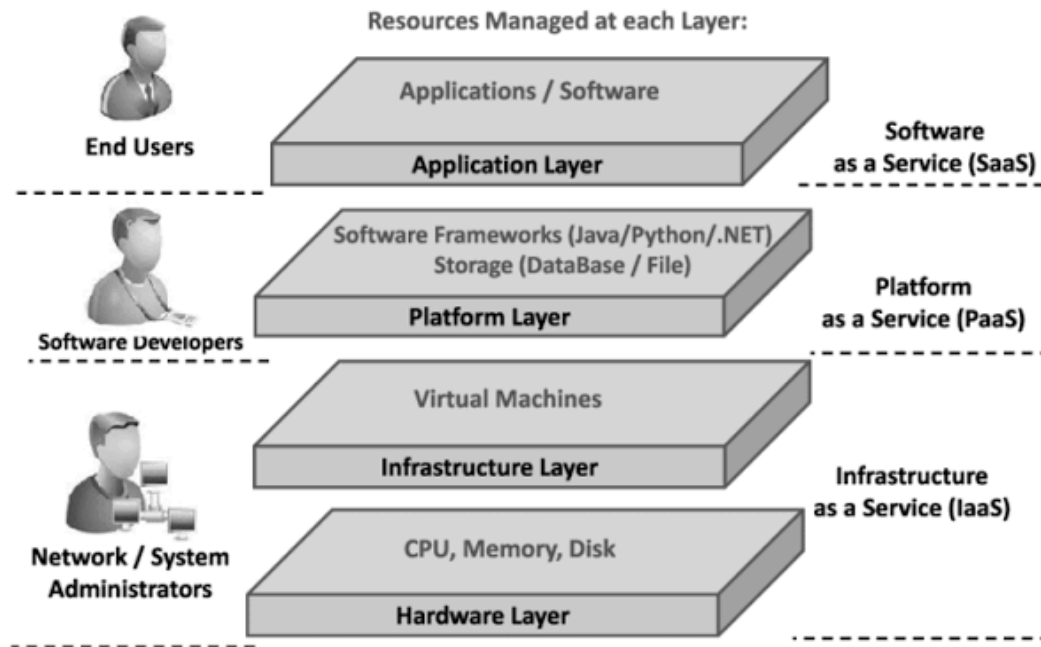
server pooling in cloud computing? Resource pooling is when cloud providers offer provisional and scalable services to multiple clients or customers. In other words, space and resources are pooled to serve multiple clients at one time. Depending on a client's resource consumption, usage can be set to provide more or less at any given time.

A **storage pool aggregates capacity and performance**. Volumes are created within the storage pool and can use the capacity and performance of that pool. All volumes in the same storage pool share the performance capability of the pool. A volume can use the full performance capacity of its pool.

Network pools (or interconnect pools) are **composed of different preconfigured network connectivity devices**. For example, a pool of virtual firewall devices or physical network switches can be created for redundant connectivity, load balancing, or link aggregation.

four layers of cloud architecture:- The Physical layer, the infrastructure layer, the platform layer and the application layer.

and the application layer [20] as shown in Figure 1.



Virtualization is technology that you can use to create virtual representations of servers, storage, networks, and other physical machines. Virtual software mimics the functions of physical hardware to run multiple virtual machines simultaneously on a single physical machine. **BENEFITS OF**

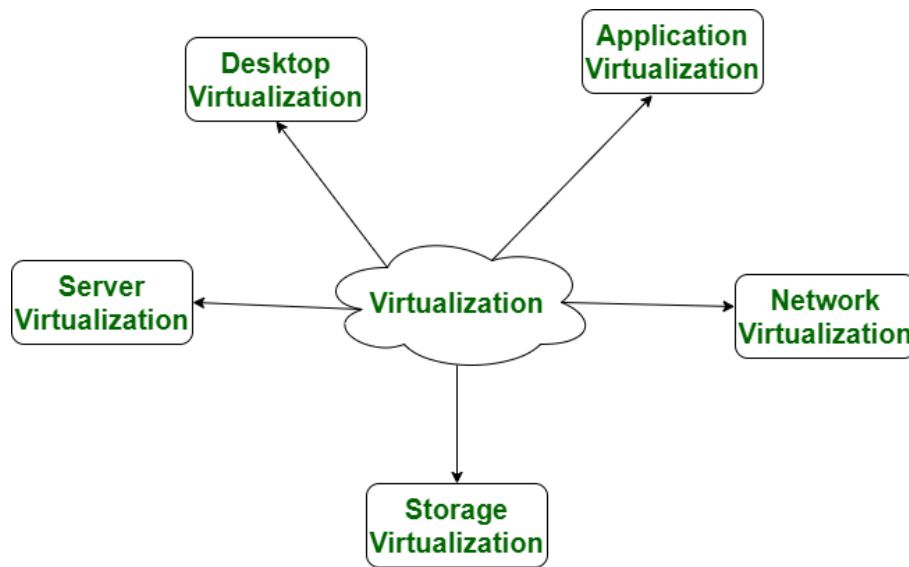
VIRTUALIZATION

1. More flexible and efficient allocation of resources.
2. Enhance development productivity.
3. It lowers the cost of IT infrastructure.
4. Remote access and rapid scalability.
5. High availability and disaster recovery.
6. Pay per use of the IT infrastructure on demand.
7. Enables running multiple operating systems.

Types of Virtualization:

1. Application Virtualization.
2. Network Virtualization.
3. Desktop Virtualization.
4. Storage Virtualization.
5. Server Virtualization.

6.Data virtualization.



Benefits of Virtualization

- Reduced upfront hardware and continuing operating costs.
- Minimized or eliminated downtime.
- Increased IT productivity and responsiveness.
- Greater business continuity and disaster recovery response.
- Simplified data center management.
- Faster provisioning of applications and resources.

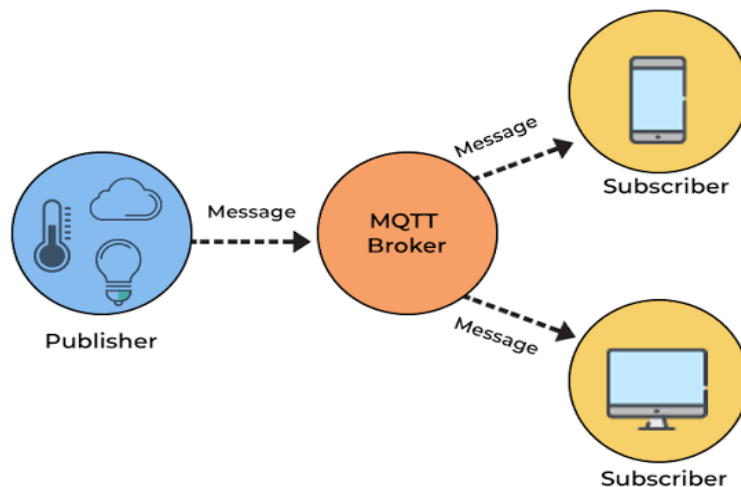
Abstraction makes it possible to encapsulate the physical implementation so that the technical details may be concealed from the customers. Virtualization makes it possible to create a virtual representation of anything, which may include computer resources, a virtual computer hardware platform, or storage devices.

IPv6 In IoT has a **highly efficient multicast communication feature that eliminates the requirement for routine broadcast messaging**. This improvement helps in preserving the battery life of IoT devices by reducing the number of packets processed. IPv6 provides multiple addresses to devices. IPv6 (Internet Protocol version 6) is the sixth revision to the Internet Protocol and the successor to IPv4. It functions similarly to IPv4 in that it **provides the unique IP addresses necessary for Internet-enabled devices to communicate**.

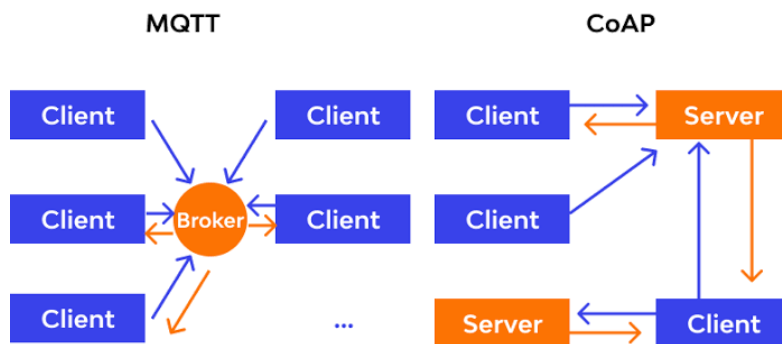
MQTT- Message queuing telemetry support (MQTT) is defined as a **low bandwidth consumption machine-to-machine protocol that helps IoT devices communicate with each other, with minimal code requirements and network footprint**. MQTT stands for message queuing telemetry transport. used- The MQTT protocol is thus energy-efficient and easy to deploy for millions of devices. Connecting devices during unreliable networks: MQTT in IoT uses QoS levels **to ensure guaranteed delivery of messages to receivers, even when connections between devices are unreliable**.



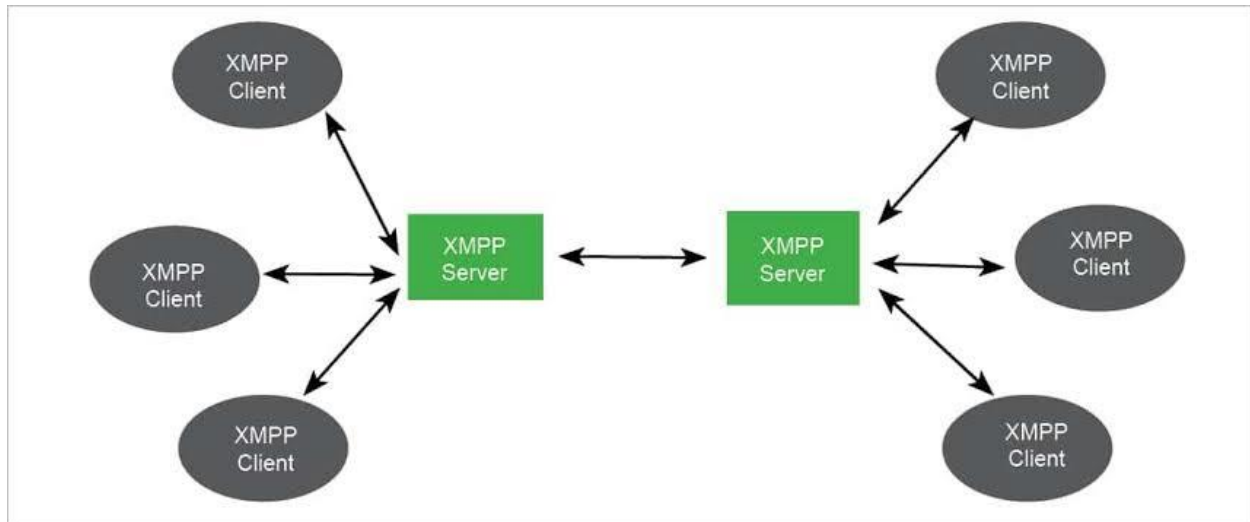
MQTT PROCESS



CoAP a customary client-server IoT protocol. It enables clients to make requests for web transfers as per the need of the hour. On the other hand, it also let supporting servers to respond to arriving requests. In summary, devices' nodes in the IoT ecosystem are enabled to interact over through CoAP only.



The **XMPP** protocol is based on the typical client server architecture, in which the XMPP client uses the XMPP server with the TCP socket. **XMPP provides a general framework for messaging across a network, offering a multitude of applications beyond traditional instant messaging (IM) and the distribution of presence data.**



- **Multi-homing** is a method of configuring one computer, called the host, with more than one network connection and IP address. The multi-homed method provides enhanced and reliable Internet connectivity without compromising efficient performance. Multi-homing is a method of configuring one computer, called the host, with more than one network connection and IP address. The multi-homed method provides enhanced and reliable Internet connectivity without compromising efficient performance. **TYPE**-Host multihoming. A single host may be connected to multiple networks. ...

- Classical multihoming. ...
- Multihoming with multiple addresses.

Horizontal scaling refers to provisioning additional servers to meet your needs, often splitting workloads between servers to limit the number of requests any individual server is getting. Horizontal scaling in cloud computing means **adding additional instances instead of moving to a larger instance size**. A simple example of horizontal scaling in AWS Cloud is **adding/removing Amazon EC2 instances from your application architecture behind Elastic Load Balancer**. Horizontal scaling (aka scaling out) refers to **adding additional nodes or machines to your infrastructure to cope with new demands**.

Challenges of Cloud Computing

- Security.
- Password Security.
- Cost Management.
- Lack of expertise.
- Internet Connectivity.
- Control or Governance.
- Compliance.
- Multiple Cloud Management.
- Creating a private Cloud
- Performance
- Migration
- Interoperability and Portability
- Reliability and High Availability
- Hybrid-Cloud Complexity